

Computer Networks (CS-241)

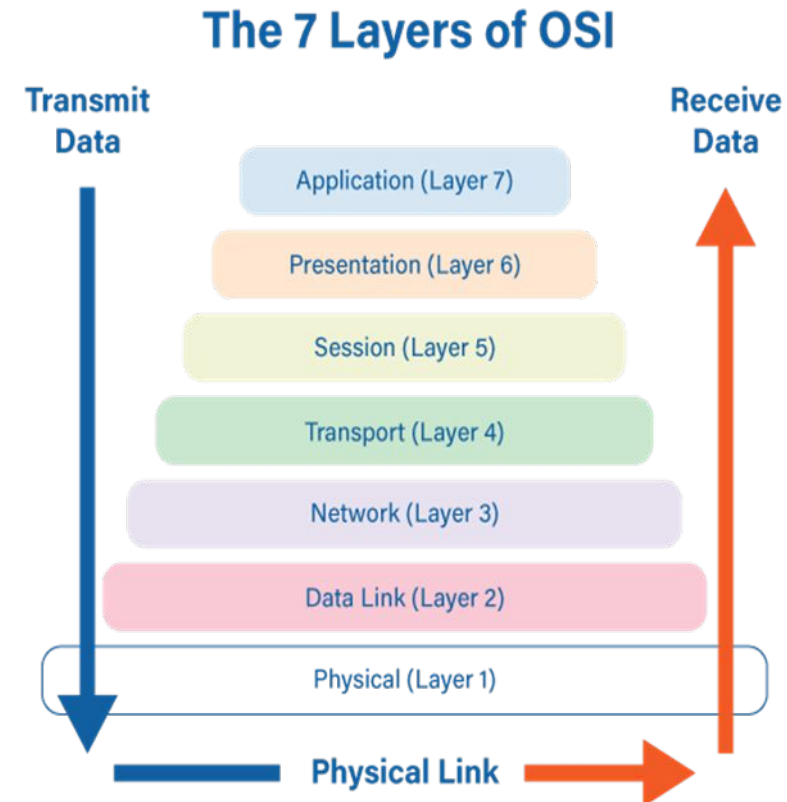
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Adapted From:

Computer Networking: A Top-Down Approach

8th edition: Jim Kurose, Keith Ross Pearson, 2020



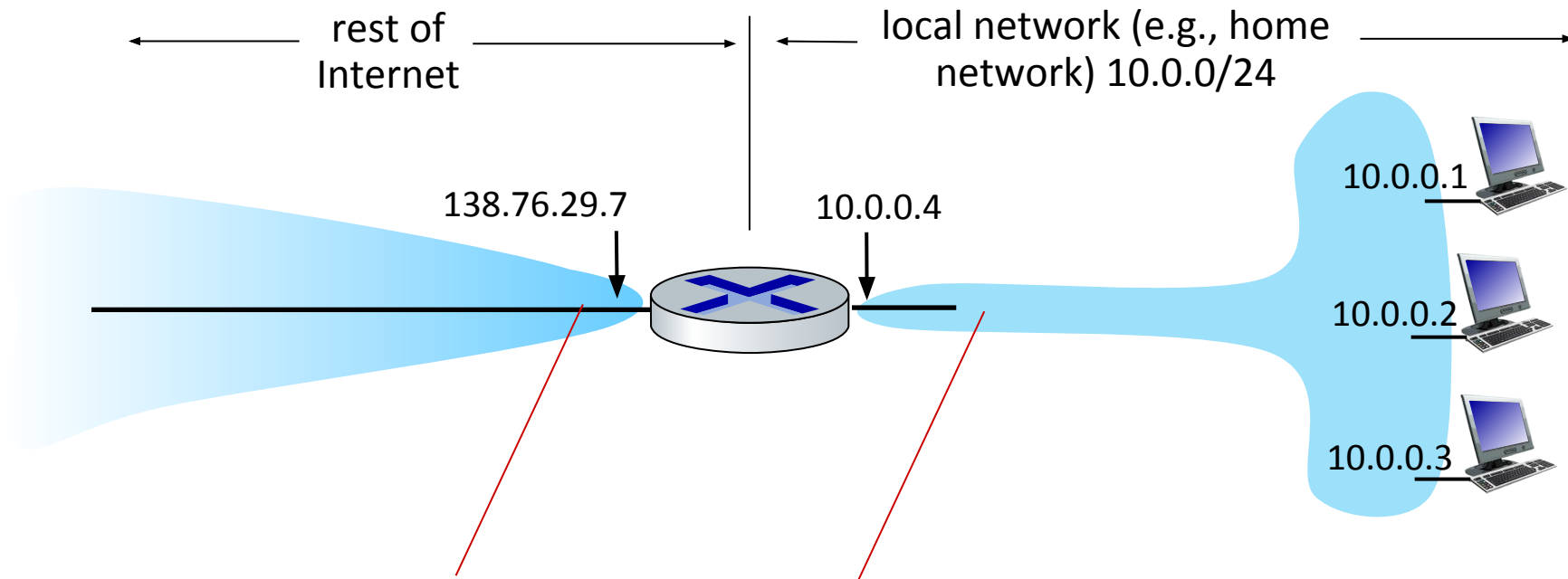
Network layer: “data plane” roadmap

- Network layer: overview
 - data plane
 - control plane
- What’s inside a router
 - input ports, switching, output ports
 - buffer management, scheduling
- IP: the Internet Protocol
 - datagram format
 - addressing
 - network address translation
 - IPv6
- Generalized Forwarding, SDN
 - match+action
 - OpenFlow: match+action in action
- Middleboxes



NAT: network address translation

NAT: all devices in local network share just **one** IPv4 address when they talk to the Internet.



all datagrams *leaving* local network have *same* source NAT IP address: **138.76.29.7**, but *different* **source port** numbers

datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

NAT: network address translation

- Every device in a local network (home, office, lab) has an IPv4 address
- An IPv4 address is 32-bit (example: 192.168.1.5). These addresses belong to private IP ranges.
- Private IPs work only inside the local network. They cannot be used directly on the Internet
- **Advantages:**
 - just **one** IP address needed from provider ISP for *all* devices
 - can change addresses of host in local network without notifying outside world
 - can change ISP without changing addresses of devices in local network
 - **security:** devices inside local net not directly addressable, visible by outside world

NAT: network address translation

Implementation: NAT router works transparently, meaning: devices don't know NAT is happening Internet also doesn't know internal IPs exist.

Step 1: Device sends data: A local device sends a packet with:

- Source IP = private IP (e.g. 10.0.0.2)
- Source Port = e.g. 5000

Step 2: NAT replaces IP and port: NAT router changes:(source IP, port) from (10.0.0.2, 5000)to (NAT public IP, new port)e.g. (138.76.29.7, 62001)

Step 3: NAT remembers the mapping:

Private IP	Private Port	Public IP	Public Port
10.0.0.2	5000	138.76.29.7	62001

NAT: network address translation

Step 4: Packet goes to Internet: Internet servers see:

- Source = **138.76.29.7 : 62001** .They reply to this address.

Step 5: Reply arrives at NAT:

- Destination IP = **138.76.29.7**
- Destination Port = **62001**

Step 6: NAT checks its table:

- NAT looks up:
 - **62001**
- Finds:
 - **10.0.0.2 : 5000**

NAT: network address translation

Step 7: NAT restores original address

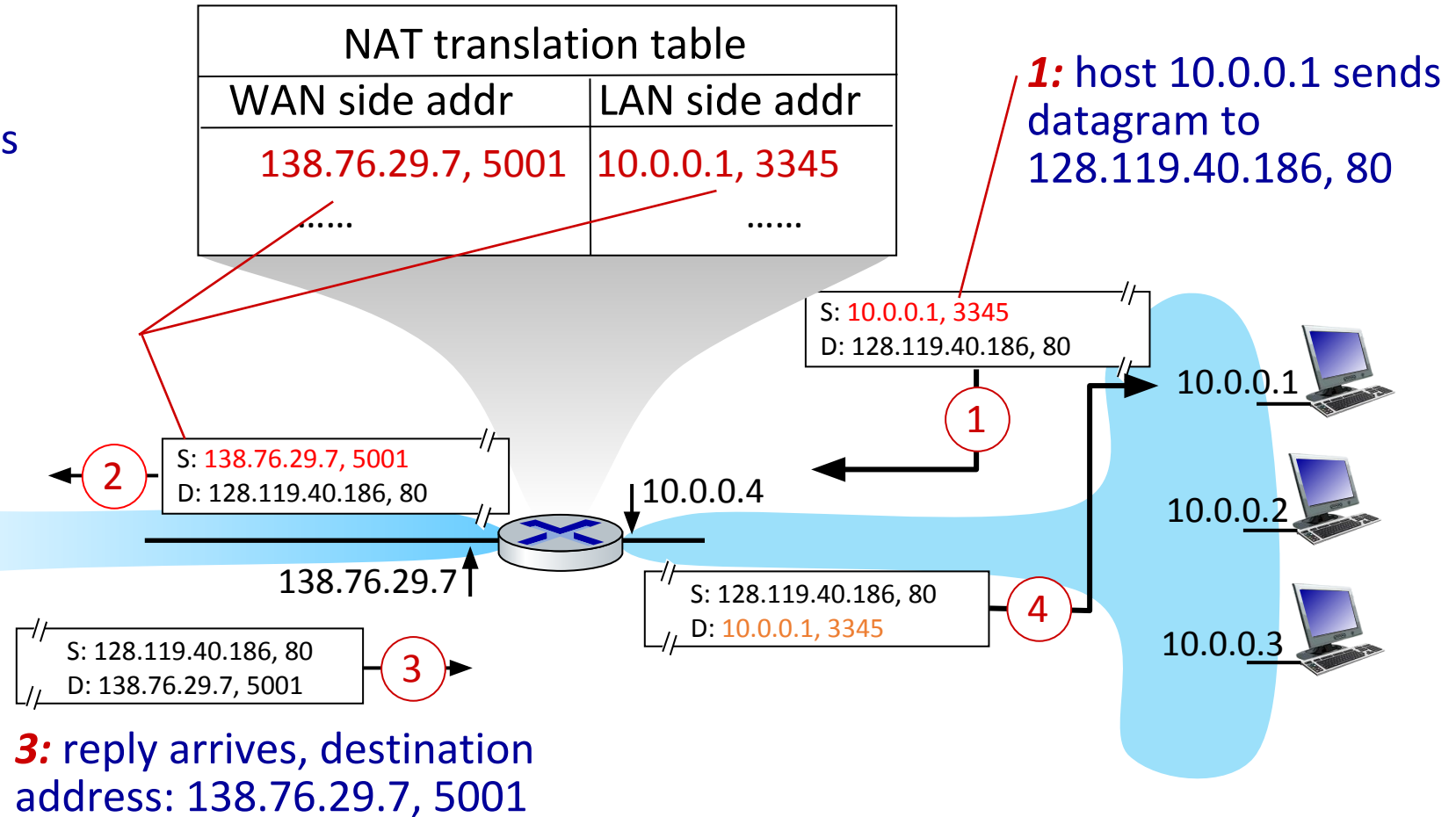
- NAT replaces destination fields:
 - from (138.76.29.7, 62001)
 - to (10.0.0.2, 5000)

Step 8: Packet delivered internally

- Packet is sent to the correct local device
- Device thinks it came directly from Internet

NAT: network address translation

2: NAT router changes datagram source address from 10.0.0.1, 3345 to 138.76.29.7, 5001, updates table



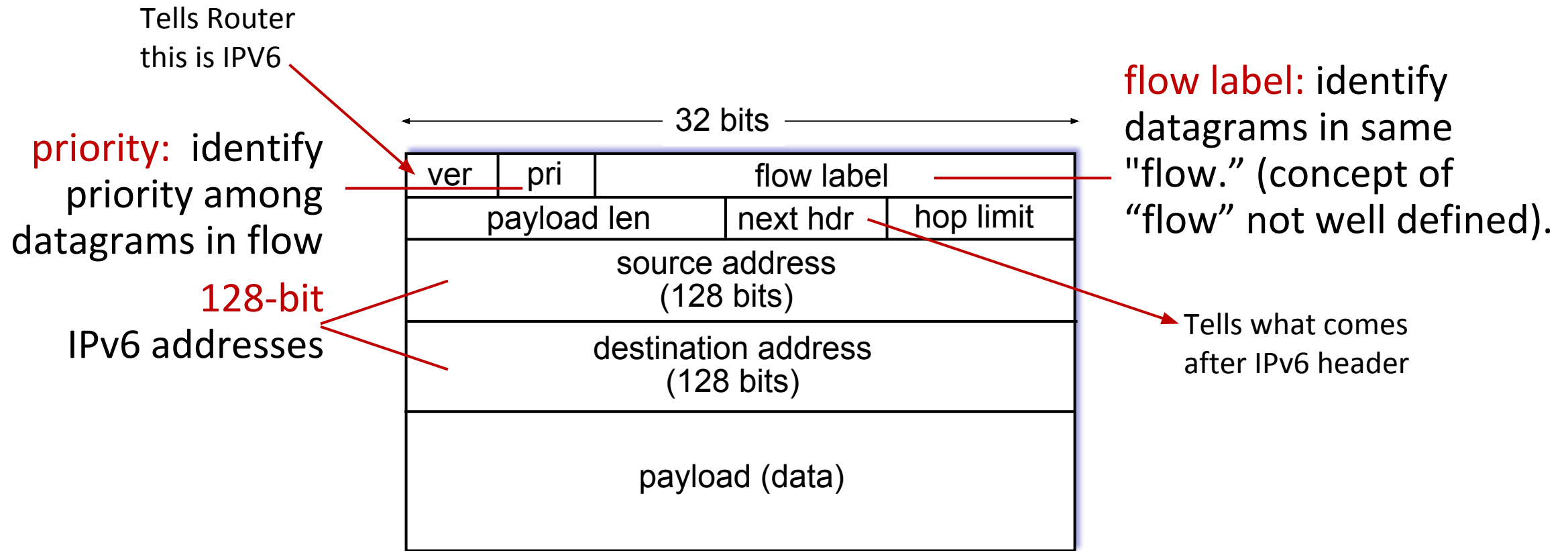
NAT: network address translation

- NAT has been controversial:
 - routers “should” only process up to layer 3
 - address “shortage” should be solved by IPv6
 - NAT problem: what if client wants to connect to server behind NAT?
 - A device behind NAT has a private IP (e.g. 192.168.1.10). This IP is not reachable from the Internet. NAT only allows incoming packets if they match an existing entry in the NAT table.
 - extensively used in home and institutional nets, **4G/5G cellular nets**

IPv6: motivation

- **initial motivation:** 32-bit IPv4 address space would be completely allocated
- additional motivation:
 - speed processing/forwarding: 40-byte fixed length header
 - Routers can handle packets more quickly because the header is simple and fixed.
 - **An IPv6 address is 128 bits long and written in hexadecimal groups separated by colons.**
 - **Examples: 2001:0db8:85a3:0000:0000:8a2e:0370:7334**
 - **Short form: 2001:db8:85a3::8a2e:370:7334**

IPv6 datagram format

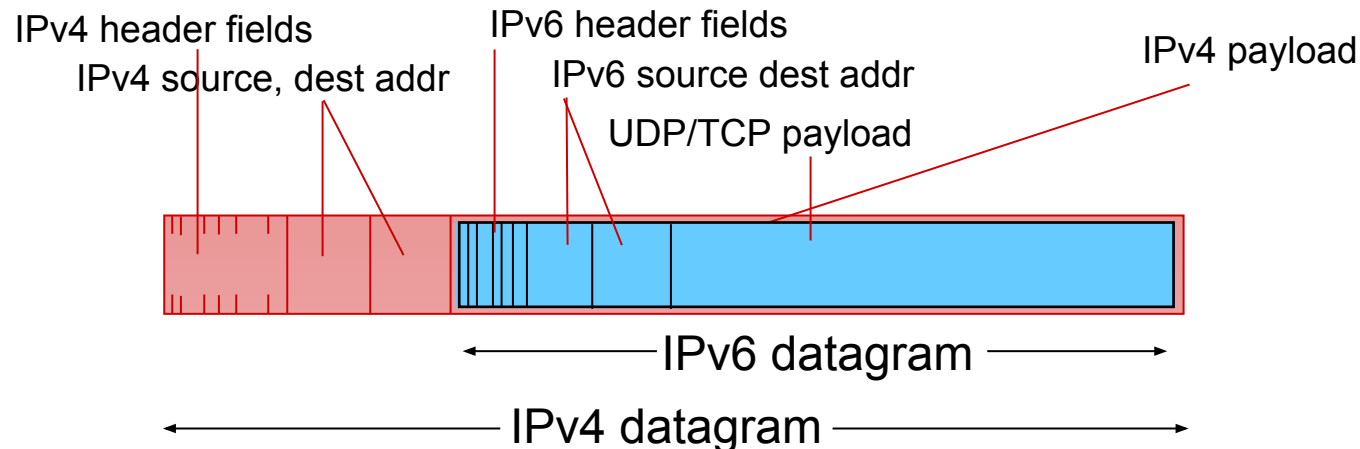


What's missing (compared with IPv4):

- no checksum (to speed processing at routers)
- no fragmentation/reassembly
- no options (available as upper-layer, next-header protocol at router)

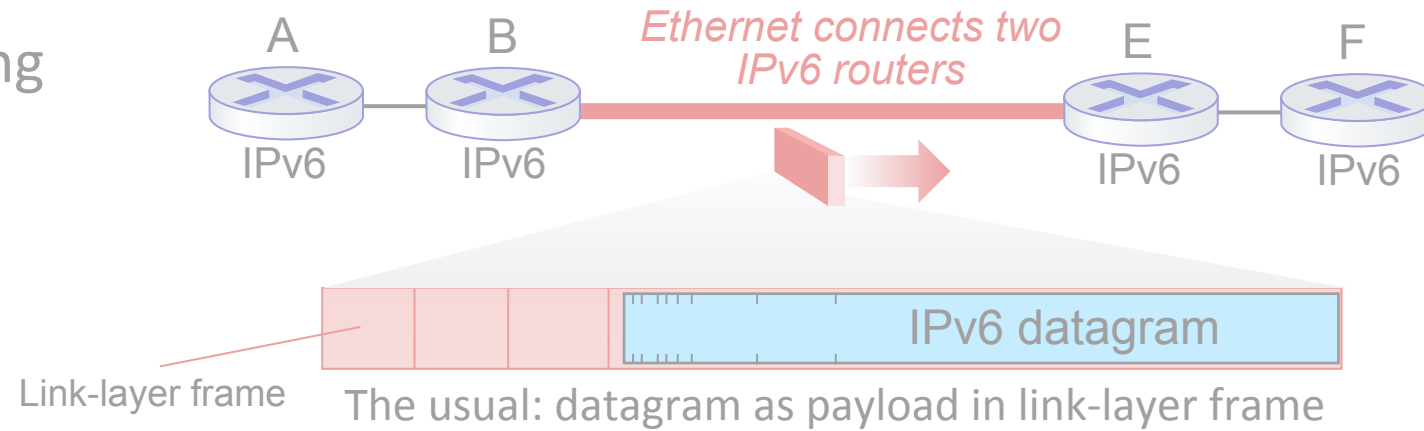
Transition from IPv4 to IPv6

- Internet has millions of routers. It is impossible to upgrade all of them at the same time. Some routers will support IPv6 Others will still use IPv4.
- **tunneling:** Tunneling carries an IPv6 datagram as payload inside an IPv4 datagram to pass through IPv4 routers, and this technique is also widely used in 4G/5G networks.

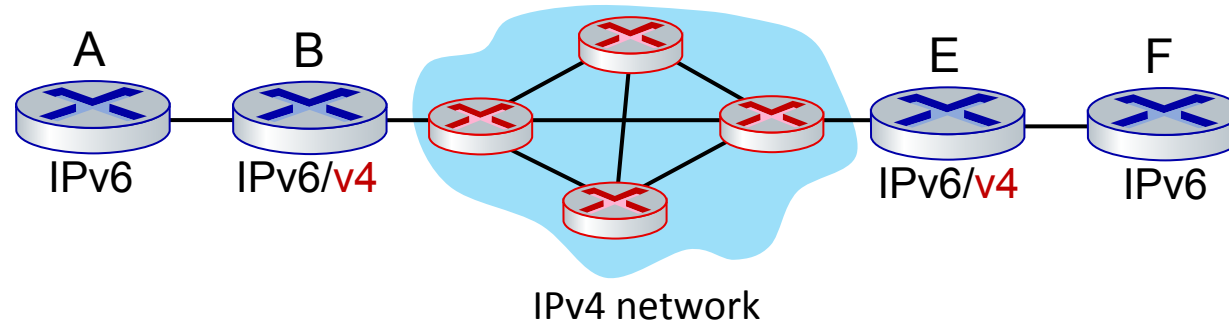


Tunneling and encapsulation

Ethernet connecting two IPv6 routers:

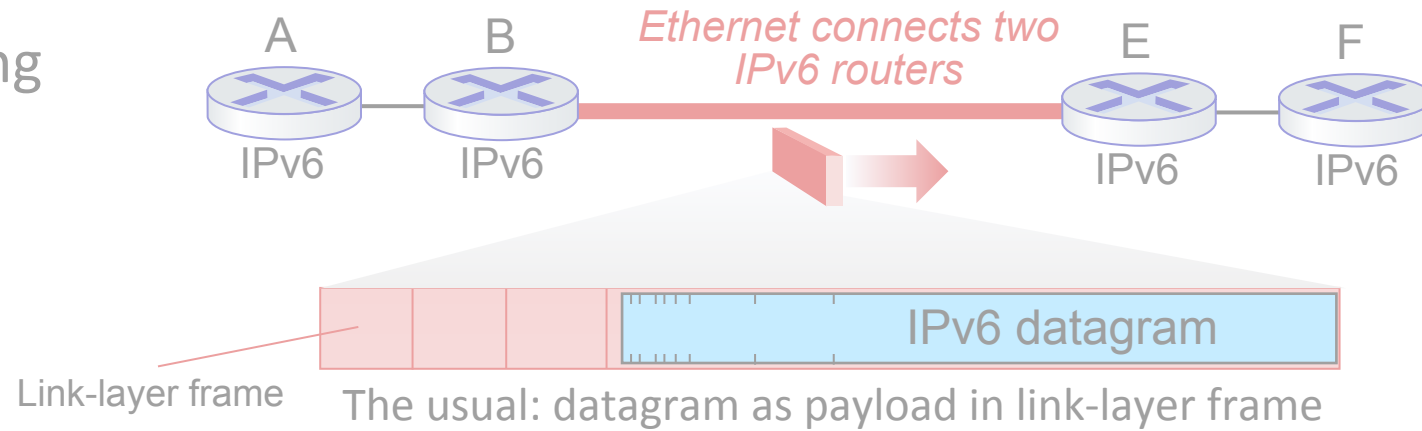


IPv4 network connecting two IPv6 routers

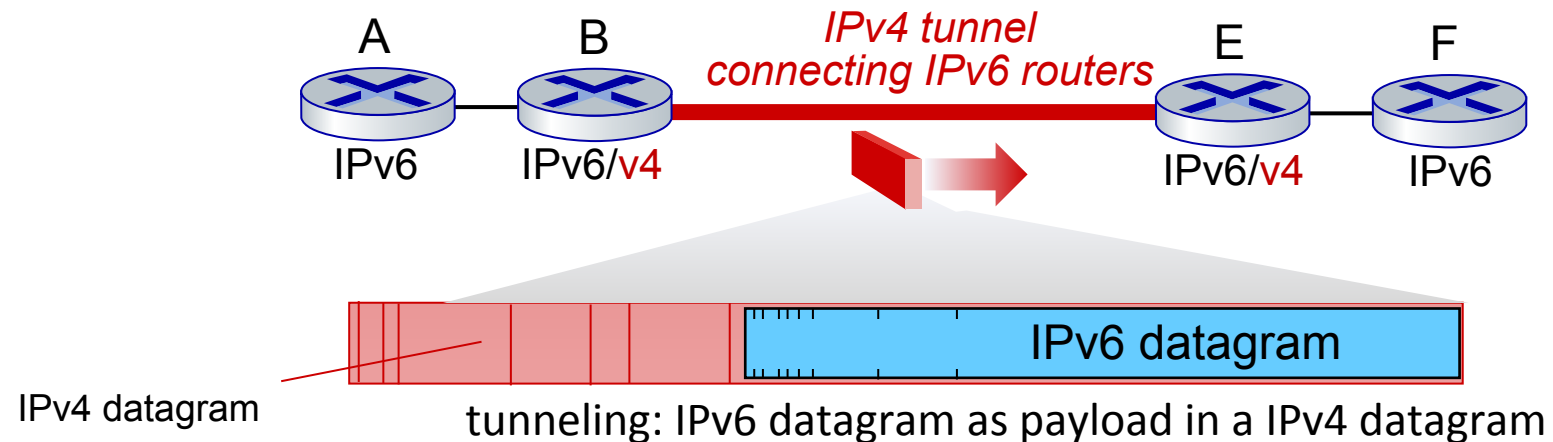


Tunneling and encapsulation

Ethernet connecting two IPv6 routers:



IPv4 tunnel connecting two IPv6 routers

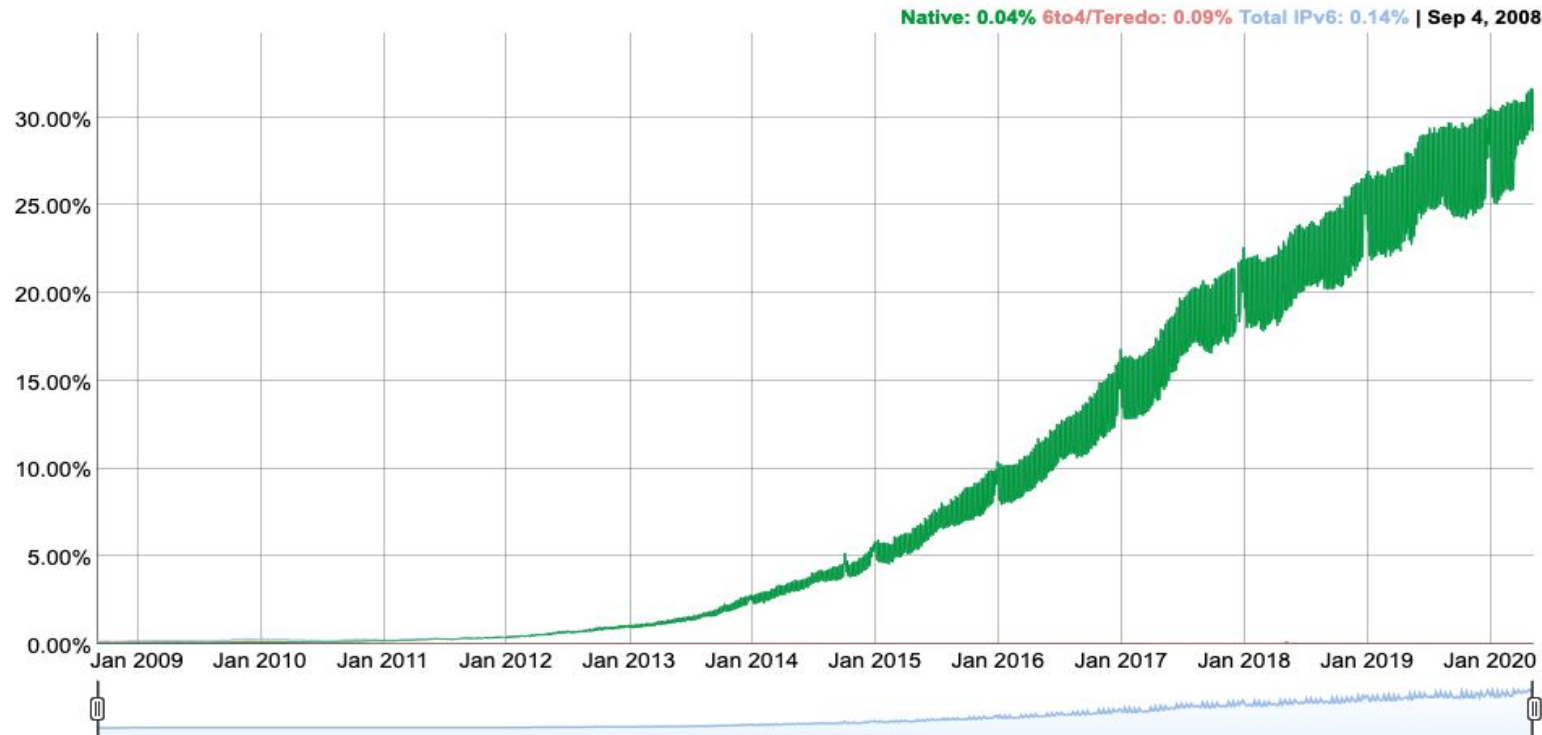


IPv6: adoption

- About 30 out of every 100 users. Use IPv6 when accessing Google services. Shows that IPv6 is actively being used, not just theoretical
- 1 out of every 3 U.S. government websites Can work with IPv6

IPv6 Adoption

We are continuously measuring the availability of IPv6 connectivity among Google users. The graph shows the percentage of users that access Google over IPv6.



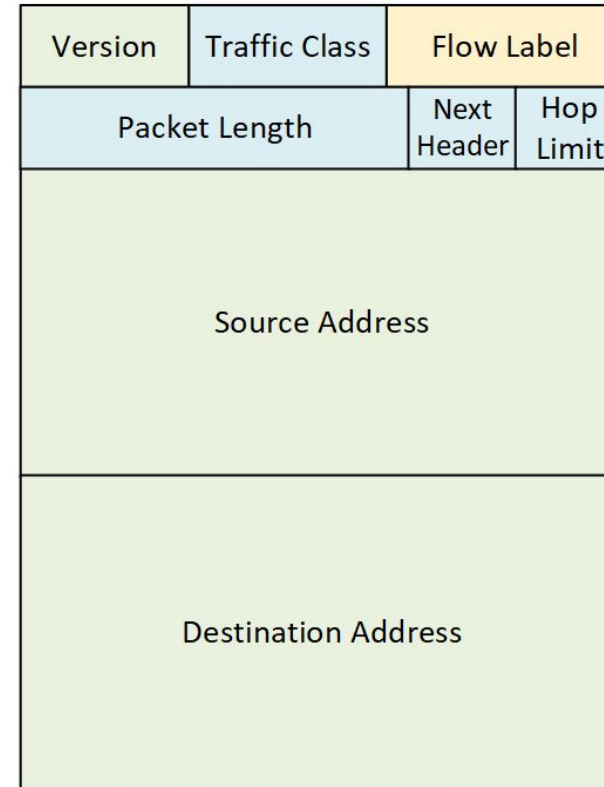
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<https://www.google.com/intl/en/ipv6/statistics.html>

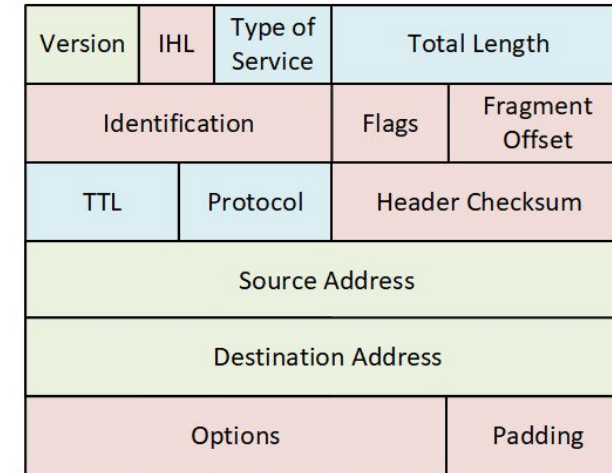
IPv4 vs IPv6

- IPv6 overcomes the limitations of IPv4 by offering scalability, better security, and simpler network management, making it essential for modern networks.

IPv6 Header



IPv4 Header



Legend

-  Fields **kept** in IPv6
-  Fields **kept** in IPv6, but name and position changed
-  Fields **not kept** in IPv6
-  Fields that are **new** in IPv6

Feature	IPv4	IPv6
Address length	32-bit	128-bit
Address exhaustion	Yes	No
Header size	Variable (20–60 bytes)	Fixed (40 bytes)
Configuration	Manual / DHCP	Auto-configuration
Security	IPsec optional	IPsec built-in
Broadcast	Supported	Not supported (uses multicast)
Performance	Lower efficiency	Improved routing efficiency